Archimedean Spiral Array With Vertically Integrated Klopfenstein and Exponential Baluns

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***Abstract*—This paper presents the design, analysis, and experimental characterization of two 3×3 Archimedean spiral antenna arrays using vertically integrated Klopfenstein and exponential tapered balun feeds. Both configurations are evaluated using full-wave simulations in Ansys HFSS and measurements of the fabricated array prototypes. A comparative study assesses input matching and interelement isolation, together with element patterns of the arrays. The results demonstrate broadband operation and quantify the influence of balun taper profile on overall array element performance due to edge effects.**

I. Introduction

Ultra-wideband (UWB) antenna arrays are a crucial feature of modern communication systems. Their usable bandwidth, however, is constrained by mutual coupling, scanning range, impedance, and pattern stability over the operating frequency range [1]. For spiral arrays, broadband performance depends not only on the radiating element but also on the feed network. A narrowband or poorly balanced transition can limit impedance bandwidth and degrade the radiation characteristics. Consequently, a broadband balun is required to maintain balanced excitation and impedance matching throughout the operating band, motivating the tapered feed networks investigated in this work. The design process and performance of two 3×3 broadband antenna arrays are presented and the influence of the baluns is studied in the context of input matching, isolation, and element patterns.

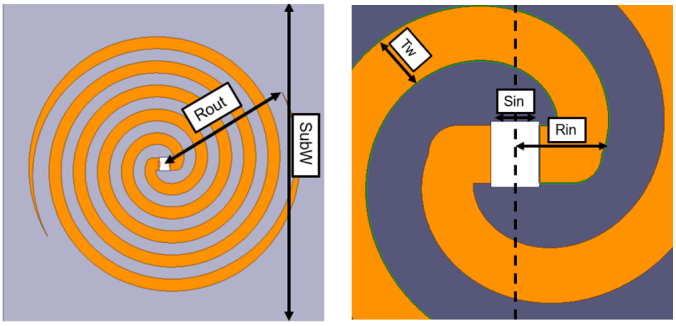
II. Design

A. Archimedean Spiral

The radiating element was implemented as a two-arm Archimedean spiral [2] with inner radius $R_{\text{in}} = 1.08$ mm, outer radius $R_{\text{out}} = 23.87$ mm and angular trace width $T_w = 2.05$ rad. Counterclockwise winding produces right-hand circular polarization under the adopted viewing convention. A gradual arm taper provides a smooth impedance transition. Parametric sweeps of R_{in}, R_{out} and T_w were conducted in Ansys HFSS [3] to balance bandwidth and input impedance. Reducing R_{in} lowered the input impedance and extended the upper operating-frequency limit. The final geometry, shown in Fig. 1, presents an input impedance of approximately 150 Ω.

B. Feed Network

The baluns were designed to transform a 50 Ω unbalanced input to the 150 Ω balanced spiral impedance while targeting $|S_{11}| < -20$ dB across the operational bandwidth. The exponential and Klopfenstein impedance profiles were synthesized using established analytical formulations [4]–[6]



(a) (b)

Fig. 1. Spiral element: (a) geometry, and (b) feed-region detail.

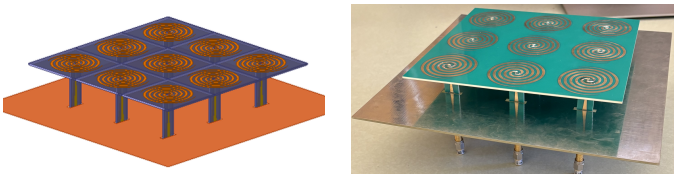
TABLE I
Taper Geometry Dimensions

Taper	Unbalanced Top Width (mm)	Unbalanced Bottom Width (mm)	Balanced Trace Width (mm)	Length (mm)
Klopfenstein	3.41	10.23	0.84	57.5
Exponential	4.00	32.00	0.84	60.0

and discretized into 11 sections. Initial conductor widths were estimated using a microstrip model. Due to the nonstandard cross-sectional geometry of the tapered balun [5], the required characteristic impedances could not be mapped directly to trace dimensions using closed-form expressions. Initial trace widths were therefore estimated using a microstrip model, after which full-wave parametric optimization was used to determine the optimal top and bottom conductor widths while maintaining a prescribed width ratio along the taper. The optimized sections were then joined using a B-spline to form smooth transition between each discretized section. The final taper dimensions are summarized in Table I. A 4-mm launch was added at the unbalanced port to accommodate an end-launch SMA connector. Both baluns and the spiral antenna were fabricated on Rogers RO4350B substrate with $\epsilon_r = 3.48$ and $\tan\delta = 0.0037$.

C. Array Configuration

The 3×3 simulation model and fabricated prototype of the spiral array are shown in Fig. 2. For brevity only the Klopfenstein tapered array is shown here. The baluns were inserted vertically into the spiral antenna geometry through a slot made in the center of each spiral antenna element, denoted by S_{in} in Fig. 1, and an additional slot in the back plane. The 3×3 array consists of identical elements arranged on a uniform square lattice with an inter-element spacing of 54.6



(a) (b)

Fig. 2. Klopfenstien tapered array: (a) simulation model, and (b) fabricated prototype.

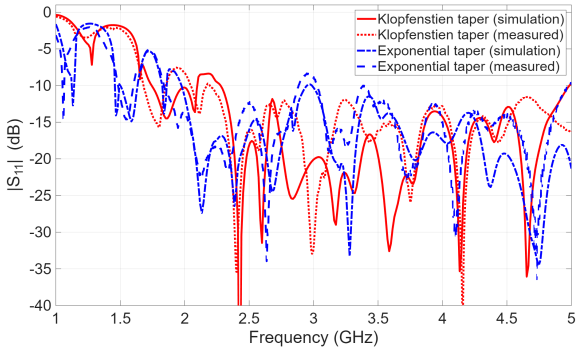


Fig. 3. Simulated and measured $|S_{11}|$ for the center-element Klopfenstein and exponential tapered arrays.

mm. The array was back-planed with a dielectric-mounted copper sheet offset approximately a quarter wavelength at the lowest operating frequency.

III. Simulated and Measured Results

Using the $|S_{11}| \leq -10$ dB criterion, the measured Klopfenstein tapered array achieves an impedance bandwidth of approximately 2.1–5.00 GHz. The exponential array exhibits slight mid-band degradation; however, it meets the criterion over 2.00–2.89 GHz and 3.02–5.00 GHz. Overall, the measured results agree well with simulation and confirm broadband impedance matching across the operating band. Interelement coupling was evaluated between the center element and the adjacent edge and corner elements, as depicted in Figure 4. Across 2–5 GHz, the band-averaged center-to-edge and center-to-corner isolation values were 18.9 and 22.4 dB for the Klopfenstein tapered array and 21.6 and 23.6 dB for the exponential tapered array.

Normalized simulated patterns are compared in Fig. 5. Both feed configurations produce comparable main-beam performance at the center frequency of the investigated bandwidth. However, their performance at the corner element shows some difference. In particular the Klopfenstein tapered array shows a beam tilt due to the edge effects which illustrates the impact of feed structure on the array pattern.

IV. Conclusions

This paper presented the design and characterization of two 3×3 Archimedean spiral antenna arrays employing vertically integrated Klopfenstein and exponential tapered baluns. Measurements showed that the Klopfenstein array shows a better performance in terms of impedance matching, while

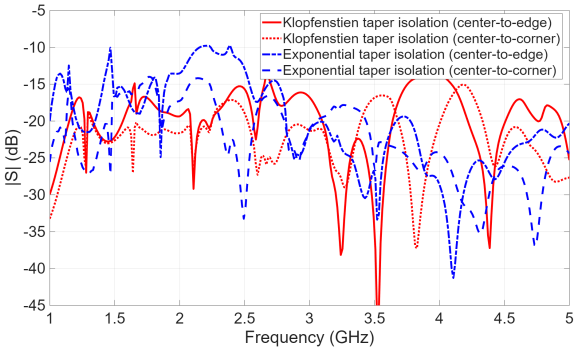
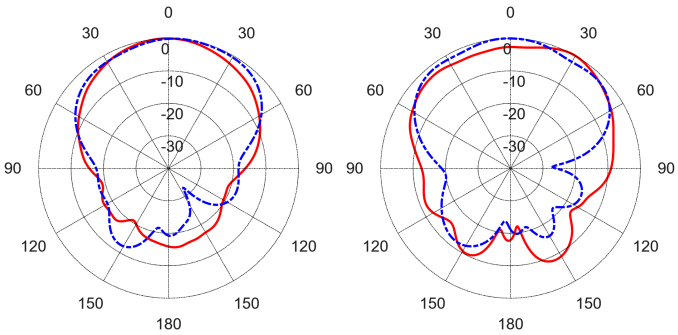


Fig. 4. Simulated center-to-edge and center-to-corner isolation for the Klopfenstein and exponential tapered arrays.



(a) (b)

Fig. 5. Normalized radiation pattern of the array elements at the center frequency band: (a) center element, and (b) corner element. The solid red and dashed blue curves correspond to the Klopfenstein and exponential tapered arrays, respectively.

the exponential array provided slightly greater interelement isolation. While both arrays achieved a comparable pattern shape at the center frequency, there are noticeable differences at corner and edge element patterns. The study reveals the importance of small scale array modeling to characterize larger arrays due to the edge effects. While the center element pattern can be computed using an infinite array approach; accurate large array modeling would require finite array simulations. These edge element effects would be further impacted by beam scanning.

References

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